Does Growth in the Money Supply Predict U.S. Inflation? Yes, but …

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KEY TAKEAWAYS

- A famous quote by Milton Friedman — that "inflation is always and everywhere a monetary phenomenon" — suggests growth in the money supply may be a reliable predictor of inflation.
- An analysis shows that money growth and inflation tend to move together mainly in extreme, high-inflation periods, but otherwise tend to drift apart.
- This is related to the velocity of the money supply, or the rate at which it circulates. At ordinary levels of money growth, changes in velocity growth can offset changes in money supply growth, nullifying the effect of excess money growth on inflation.
- Thus, while money growth seems to be a poor everyday predictor of inflation, it does seem to provide a useful signal when the money growth rate moves far outside its historical range.

Milton Friedman famously said, "Inflation is always and everywhere a monetary phenomenon in the sense that it is and can be produced only by a more rapid increase in the quantity of money than in output."[1]

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A natural question, then, is whether money supply growth is a reliable predictor of inflation.

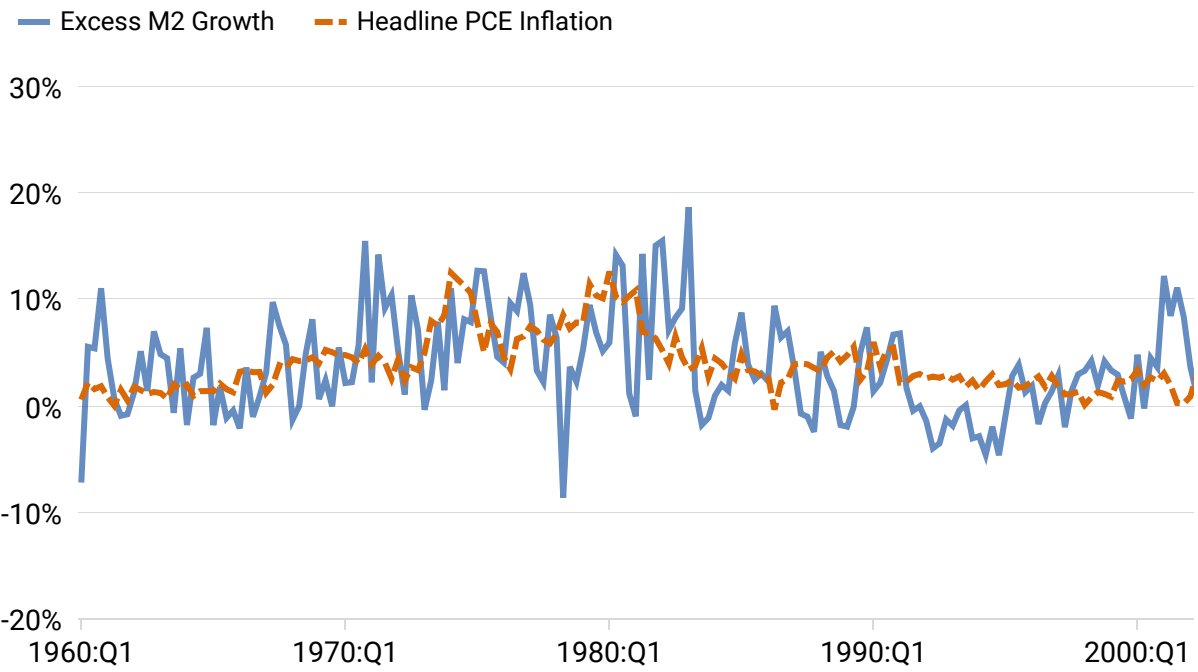
Examining the Evidence

Evidence in favor of Friedman's claim that inflation is a monetary phenomenon can be found in the first figure below. Here we plot quarterly values of excess money growth, which is annualized growth in the supply of money (M2) in excess of growth of the U.S. economy (Friedman's "output") as the solid blue line. With the dashed orange line, we plot headline personal consumption expenditures (PCE) inflation, which is the Federal Reserve's preferred inflation measure.

It is clear that inflation began to rise sharply in 2021 and into 2022 following a rapid rise in money growth that began in early 2020. At its peak in mid-2020, excess money growth reached nearly 93%, unmatched anywhere else in the 66-year sample, and PCE inflation followed roughly a year later, eventually peaking at more than 7% in mid-2022.

Excess Money Growth and U.S. Inflation

First Quarter 1960 to Second Quarter 2026



SOURCES: Federal Reserve Board (H.6), U.S. Bureau of Economic Analysis (NIPA), Haver Analytics and authors' calcula
NOTES: This analysis uses quarterly data. Excess M2 growth is calculated as annualized M2 growth minus annualized growth in the personal consumption expenditures price index. Excess M2 growth reached 92.5% in 2020:Q2; to make t from the figure.

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And yet the figure also presents evidence that is less consistent with Friedman's claim. Across most of the sample, growth in the money supply has been greater than growth of real gross domestic product (GDP) in the U.S. Nevertheless, it has

been very rare that inflation has consistently veered above the Federal Reserve's inflation target of 2%, especially since the early 1990s. It is for this reason that economic forecasters rarely consider growth in the money supply as an important indicator of future movements in inflation. In fact, a well-respected academic monograph on forecasting inflation doesn't reference money growth even once.[2]

This dichotomy is discussed in the 2023 Bank for International Settlements (BIS) article by Claudio Borio, Boris Hofmann and Egon Zakrajšek. They investigate the link between excess money growth and inflation across 32 countries, including the U.S., using non-overlapping 10-year averages. They conclude that there is a near perfect one-to-one relationship between the two variables when inflation is high, but that the relationship is significantly weaker at low to modest levels of inflation. This aligns well with the first figure, which shows that money growth and inflation move together during the rare episodes when inflation is elevated, but drift apart the rest of the time.

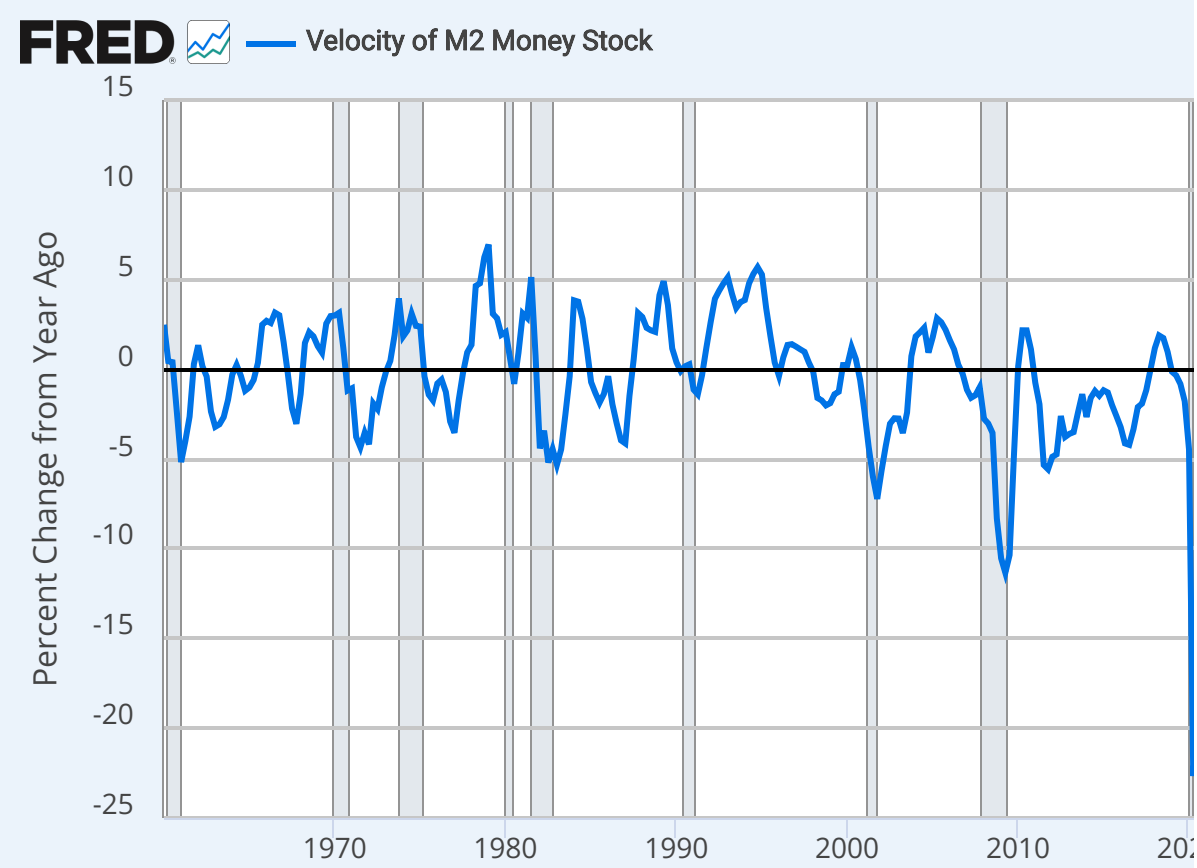
Velocity and the Quantity Theory of Money

To understand why this happens, we must bring another term into the equation: the velocity of money supply. Friedman's quote is inextricably linked to the quantity theory of money, which holds that the money supply (M) multiplied by the rate at which it circulates (velocity, V) equals nominal income (price level P multiplied by real income Y): MV = PY. This relationship is static, applying only at a moment in time, whereas inflation occurs across time.

In growth-rate terms, the growth rate of money + the growth rate of velocity = inflation + the growth rate of income. Rearranging, excess money growth = inflation - velocity growth — almost exactly Friedman's quote, except that velocity growth is now involved. That additional component can matter: If velocity growth decreases as excess money growth increases, the effect of excess money growth on inflation may be nullified.

The next figure plots the growth rate of M2 velocity — nominal GDP divided by the same M2 stock used in the first figure — over the same 66-year span. Velocity growth is anything but a constant near zero. It climbed sharply in the early 1990s, just as excess money growth turned negative, and declined substantially in the early 2000s as excess money growth picked back up.

Its widest swings, though, came during the two episodes when the money-inflation link was tested most severely: Velocity growth collapsed to roughly –11% during the 2008-09 financial crisis even as excess money growth surged into double digits, and it swung from about –23% in mid-2020 to +11% by 2023 as the pandemic-era money stock first exploded and then contracted. In both cases, offsetting movements in velocity absorbed much of what a naive reading of the quantity equation would have attributed to future inflation.



Source: Federal Reserve Bank of St. Louis via FRED®

Shaded areas indicate U.S. recessions.

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Those two episodes explain why the money-inflation link is unreliable as a routine forecasting tool. In 2008-09, double-digit excess money growth coincided with falling inflation because velocity fell faster than money grew — banks and households hoarded the new money rather than spending it. In 2020-21, by contrast, money growth was so extreme, and so directly tied to transfers into household bank accounts, that even a sizable offsetting decline in velocity was not enough to prevent the subsequent inflation surge.

This is consistent with Borio, Hofmann and Zakrajšek's finding that money growth carries information about inflation mainly in extreme, high-inflation episodes: It is precisely when the shock is large enough to overwhelm velocity's usual offsetting behavior that Friedman's rule of thumb holds. At ordinary levels of money growth, velocity fluctuations swamp the signal, which is exactly why forecasters have largely set money growth aside.

Conclusion

Taken together, it is not at all obvious that growth in the money supply is a reliable, everyday predictor of inflation like one might expect from Friedman's quote. And yet, neither is it irrelevant. Across much of the post-1960 sample, persistent excess money growth coexisted with low and stable inflation. Only in rare episodes when money growth turned extreme — the 1970s and again in 2020-21 — did the quantity-

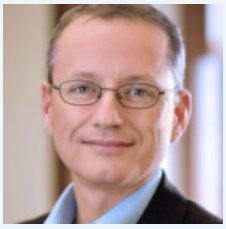
theory relationship reassert itself. For a forecaster, that makes money growth a poor everyday predictor, but a useful early-warning signal precisely when it matters most: when its growth rate moves far outside its historical range.

Notes

1. See Milton Friedman's 1970 Wincott Memorial Lecture, "The Counter-Revolution in Monetary Theory." It was also published as Institute of Economic Affairs Occasional Paper No. 33.

2. See Jon Faust and Jonathan H. Wright's chapter, "Forecasting Inflation," in the 2013 book *Handbook of Economic Forecasting: Volume 2A*.

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